

Product Requirements Document (PRD)

AI-Powered Timesheet Approval & Invoice Management Platform

Company: Lorvish Technologies

Version: 1.0

Status: Draft

1. Product Overview

The AI-Powered Timesheet Approval & Invoice Management Platform automates the complete timesheet approval and invoice generation process while maintaining required human approvals. The platform simplifies operations for staffing and consulting businesses by reducing manual work, improving billing accuracy, and providing complete visibility across the approval lifecycle.

2. Objectives

- Automate the timesheet approval process.
 - Reduce manual invoice preparation.
 - Improve billing accuracy.
 - Speed up invoice generation.
 - Automate email communication.
 - Maintain complete audit history.
 - Provide dashboards for all stakeholders.
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3. Business Workflow

Step 1 – Timesheet Submission

- Candidate submits a timesheet.
 - System validates the submitted data.
 - Timesheet status becomes **Submitted**.
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Step 2 – Client Manager Approval

- System sends an approval request via Microsoft Graph API.

- Client Manager can:
 - Approve
 - Reject
 - Add comments

If Rejected

- Candidate receives rejection reason.
- HR Team is notified.
- Candidate edits and resubmits the timesheet.

If Approved

- Timesheet moves to HR and Accounts for validation.
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Step 3 – HR & Accounts Validation

HR verifies:

- Employee allocation
- Attendance
- Contract details

Accounts verifies:

- Billable hours
- Billing rate
- Client information

After validation, the workflow proceeds to invoice generation.

Step 4 – AI Invoice Generation

The AI engine automatically:

- Calculates billing amount
- Calculates taxes
- Generates invoice
- Creates invoice PDF
- Drafts invoice email

Step 5 – Accounts Review

Accounts Team reviews:

- Invoice details
- Billing calculations
- Tax calculations
- Invoice PDF
- Email draft

They can:

- Approve
- Reject
- Edit before sending

Step 6 – Invoice Delivery

- Approved invoice is sent to the client using Microsoft Graph API.
- Invoice status is updated.

Step 7 – Tracking & Audit

The system maintains:

- Invoice status
- Approval history
- User activity
- Audit logs

4. User Roles

Candidate / Employee

- Submit timesheets
- View status
- Edit rejected submissions

- Resubmit timesheets
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Client Manager

- Review assigned timesheets
 - Approve or reject
 - Add comments
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HR Team

- Validate approved timesheets
 - Review employee details
 - Approve or reject
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Accounts Team

- Validate billing
 - Review AI-generated invoice
 - Edit invoice
 - Approve invoice
 - Send invoice
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Client

- Receive invoices
 - Receive email notifications
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System Administrator

- Manage users
- Manage roles
- Configure workflows
- View audit logs

- Manage system settings
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5. Core Modules

Authentication & User Management

- Login
 - JWT Authentication
 - Role-Based Access Control (RBAC)
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Timesheet Management

- Submit timesheet
 - Edit timesheet
 - Resubmit rejected timesheet
 - Track submission status
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Approval Workflow

- Client approval
 - Rejection workflow
 - Comments
 - Approval notifications
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HR & Accounts Validation

- Validation queue
 - Review process
 - Approval history
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AI Invoice Generation

- Billing calculation
- Tax calculation

- Invoice generation
 - PDF generation
 - Email draft generation
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Invoice Management

- Review invoice
 - Edit invoice
 - Approve invoice
 - Send invoice
 - Track invoice status
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Email Notifications

- Approval requests
 - Rejection notifications
 - Invoice delivery
 - System notifications
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Dashboard & Reports

- Timesheet summary
 - Pending approvals
 - Invoice summary
 - Reports
 - Analytics
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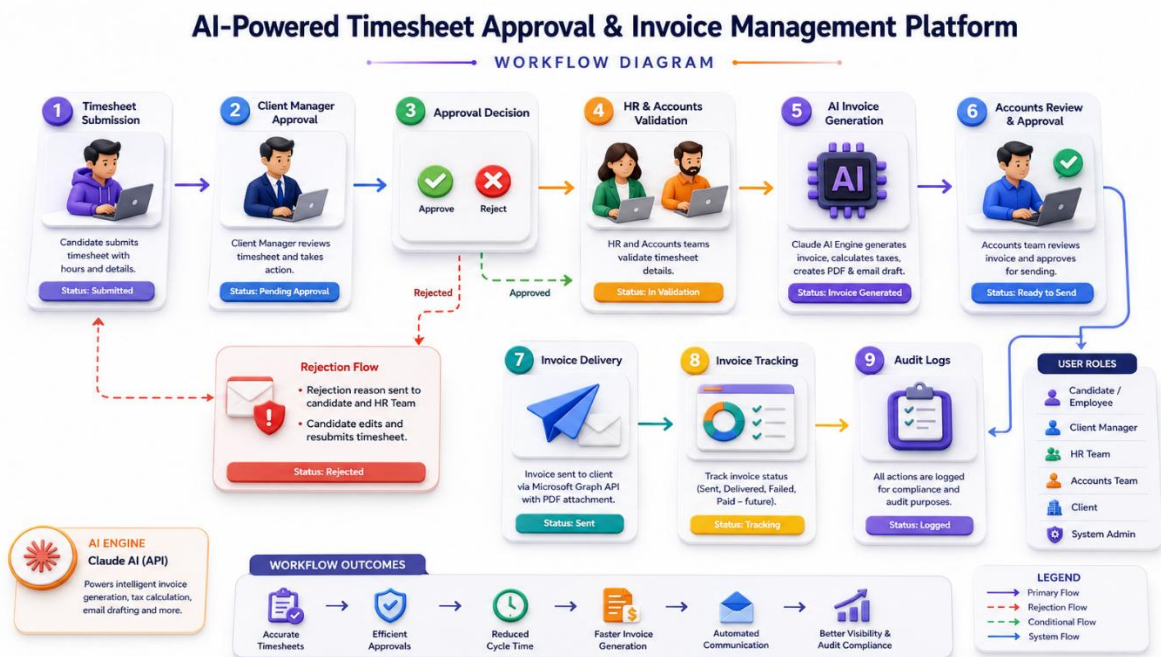
Audit Logs

- Login activity
- Approval history
- Invoice history
- User actions

6. System Architecture



7. Workflow Diagram



8. Dashboard Overview

Candidate Dashboard

- Submitted timesheets
 - Approval status
 - Rejected submissions
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Client Manager Dashboard

- Pending approvals
 - Approved timesheets
 - Rejected timesheets
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HR Dashboard

- Pending validations
 - Validation history
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Accounts Dashboard

- Pending invoices
 - Generated invoices
 - Sent invoices
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Admin Dashboard

- Users
 - Workflow statistics
 - Audit logs
 - System activity
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9. Technology Stack

Layer	Technology
Frontend	Next.js, React, TypeScript, Tailwind CSS
Backend	NestJS, Node.js or fastapi
Database	Supabase PostgreSQL
Storage	Supabase Storage
AI	OpenAI
Email Service	Microsoft Graph API
Authentication	JWT, RBAC
Phase 1 Deployment	Vercel + Railway + Supabase
Future Deployment	Azure / AWS

10. Phase 1 Scope

Included

- User Authentication
- Role-Based Access Control
- Timesheet Submission
- Client Approval Workflow
- HR Validation
- Accounts Validation
- AI Invoice Generation
- Invoice PDF Generation
- Email Draft Generation
- Microsoft Graph Integration
- Invoice Approval

- Invoice Delivery
 - Dashboard
 - Reports
 - Audit Logs
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11. Future Enhancements

- Client Portal
 - Payment Tracking
 - Payment Reminders
 - Multi-Currency Support
 - ERP Integration
 - Accounting Software Integration
 - AI Fraud Detection
 - Advanced Analytics
 - Azure Deployment
 - AWS Deployment
 - Mobile Application
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12. Success Metrics

- Faster approval process
- Reduced manual effort
- Faster invoice generation
- Improved billing accuracy
- Reduced processing errors
- Email delivery success
- User adoption
- System uptime